LIU XINCHANG (Leo)

Product & Interaction Designer · Singapore

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Product & interaction designer exploring new forms of physical interaction through materials, computation, and fabrication. I turn ideas into working prototypes, bringing together embedded electronics, mixed reality, and digital fabrication, through hands-on experimentation, testing, and iteration.

EDUCATION

National University of Singapore Aug 2025 – Nov 2026 (expected)
Research Master in Industrial Design · Soft Technologies Lab, College of Design and Engineering

Hunan University of Technology Jul 2020 – Jun 2024
Bachelor of Design, Product Design

EXPERIENCE

3D Printing & Medical Visualization Intern Apr 2026 – Oct 2026
3D Printing & Prototyping (3DPP) Lab, Duke-NUS Medical School / SingHealth

- Developed and fabricated 3D-printed anatomical models for clinical education and surgical planning, working closely with doctors, resident doctors, and medical researchers.
- Built and refined 3D models in Blender, Rhino, and SolidWorks, preparing print-ready files and optimising digital-to-physical fabrication pipelines.
- Supported silicone moulding and rapid prototyping of medical simulation devices.
- Translated clinical requirements into physical prototypes in collaboration with clinicians and researchers.
- Managed day-to-day lab operations and maintained specialised fabrication equipment; supported demonstrations, public lectures, and Open House activities on medical 3D printing and digital manufacturing.

Curator Intern · China Textile Publishing House Jun 2023

- Curated a fashion and design exhibition for about 1,500 attendees, shaping the exhibition concept and selecting fashion products and design prototypes for display.
- Managed event logistics and on-site operations, including artefact layout, venue arrangement, and stage presentation.
- Coordinated communication between designers, exhibitors, models, and the organising team to keep the event on schedule.

Interior Design Intern · Neta Auto Mar 2022

- Worked on the vehicle centre console, developing interior components with a focus on comfort, spatial ergonomics, and aesthetics, informed by user research and benchmarking.
- Validated console designs before tooling with FEA (SolidWorks Simulation, ANSYS) and mould-flow analysis (Autodesk Moldflow, SolidWorks Plastics), predicting stress, fill, weld lines, air traps, sink marks, and warpage.
- Carried out DFM checks with engineers (draft angles, wall thickness, undercuts, parting lines) so console designs were ready for tooling.
- Ran design-of-experiments on moulding process parameters, and applied root cause and failure analysis to cosmetic and dimensional defects (warpage, shrinkage, cracking, flow marks, short shots), fixing them at source.
- Designed 3D-printed jigs and fixtures for prototype assembly and inspection, defining critical tolerances with GD&T; tracked critical dimensions across batches with SPC control charts.
- Coordinated with material suppliers and mould makers on specifications, lead times, and incoming quality.
- Produced design presentations in Photoshop, Illustrator, InDesign, and Figma, including mood boards and material boards.

RESEARCH & DESIGN PROJECTS

PeraNETS ACM DIS 2026 Soft Technologies Lab, NUS
Peranakan-Inspired Architectural Tiles as Modular Interfaces

- Reimagined heritage ornament as a network of interactive, computational tiles that sense gestures and environmental change and exchange information with their neighbours.
- Fabricated layered, inlaid acrylic tiles with embedded LEDs and tile-to-tile connectors, scaling from a single tile to wall and floor surfaces.

Skills: physical computing · modular networked hardware · laser cutting · material and lighting design

GuitarIO Soft Technologies Lab, NUS

An interactive guitar for adaptive, embodied learning

- Built a guitar with a 3D-printed conductive fret array (6 strings × 23 frets, read through an analog multiplexer), per-string pressure sensing, 6 × 92 WS2812B LEDs on the neck, and vibrotactile error feedback.
- Designed a companion interface for progression, accuracy, and a real-time virtual fretboard; guidance fades as the learner improves.
- Evaluated with 25 participants (15 beginner, 10 intermediate) across feedback and learning modes.

Skills: embedded sensing · Arduino / C++ · adaptive interaction design · UI design · user studies

MetaPatch Soft Technologies Lab, NUS

NFC-based semantic interaction for user-defined mixed reality experiences

- Turned battery-free NFC stickers into rewritable "semantic patches" that let users author meaning into everyday objects and read it back as MR overlays.
- Built body-worn NFC readers for the wrist, finger, and insole (PN532 + Adafruit Feather nRF52840, BLE 5.0) linked to a Unity / OpenXR headset app with hand-tracked spatial anchors.
- Designed 3D-printed tangible buttons and sliders, and six scenarios from furniture assembly and card games to resident-authored navigation.

Skills: NFC / BLE · Unity · OpenXR · wearable hardware · tangible interface design · interaction frameworks

Water Clear: An Automated Trash-Collecting Boat Team Leader · Undergraduate

- Led task allocation and progress, wrote part of the control code, and designed the final presentation.

PUBLICATION

"PeraNETS: Peranakan-Inspired Architectural Tiles as Modular Interfaces." *ACM Designing Interactive Systems Conference (DIS 2026)*.

SKILLS

Physical computing: Arduino, C/C++, PCB design, multi-sensor fusion, NFC (PN532), BLE, Adafruit Feather nRF52840

AI & ML: LLM API integration (OpenAI / local-model APIs); supervised ML for gesture & playing-pattern classification

CAD & rendering: Rhino, SolidWorks, Blender, KeyShot

Research: user studies, prototyping, academic writing

Interactive & XR: Unity, OpenXR, mixed reality, animation, game development

Fabrication: 3D printing, laser cutting, silicone moulding, rapid prototyping

Design & presentation: Photoshop, Illustrator, InDesign, Figma, Premiere Pro, mood boards, material boards

AWARDS

Silver Award, "Create Youth" Provincial Competition (Group Leader) 2022

Third Prize, Internet+ Innovation and Entrepreneurship Competition 2021